

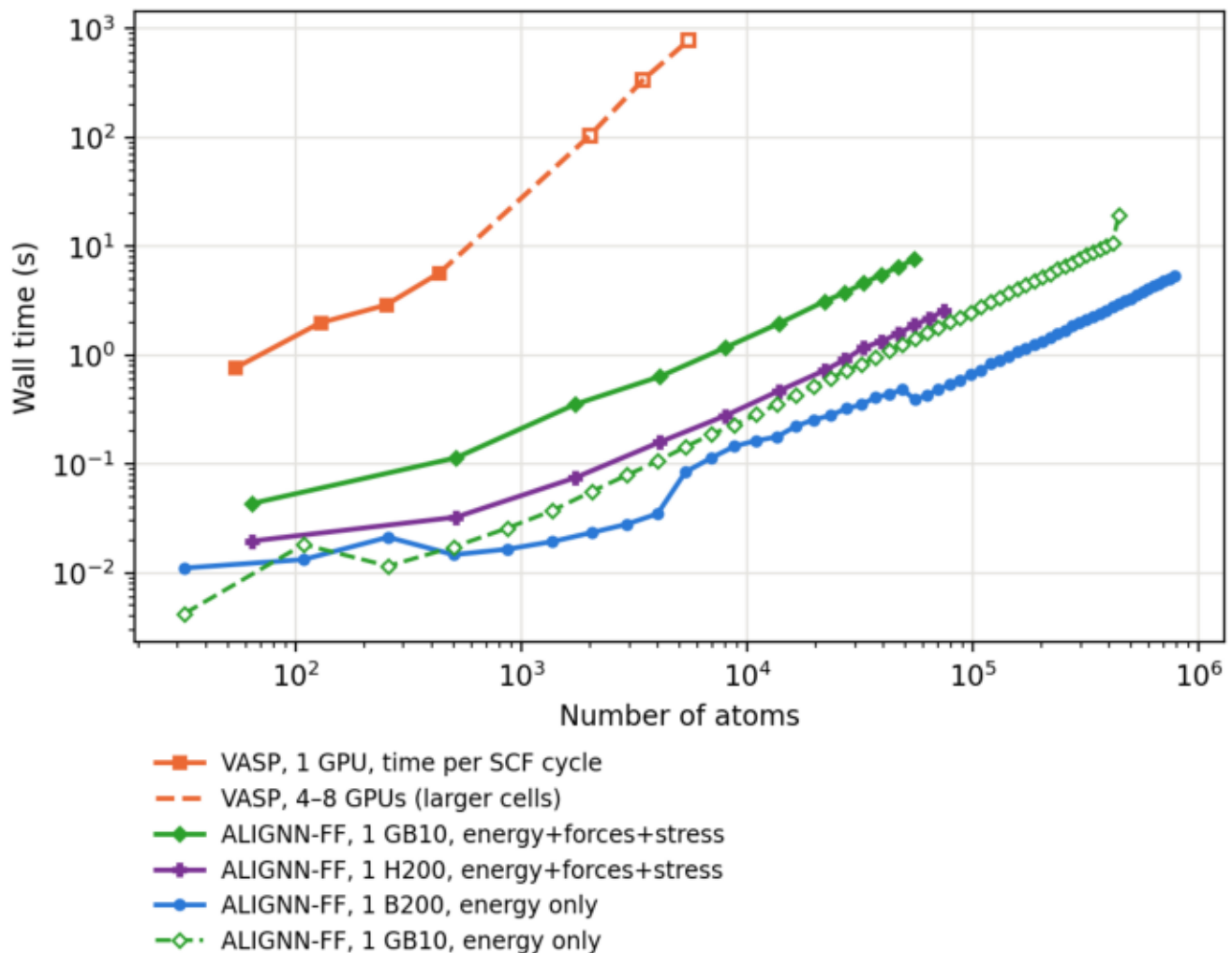
GPU scaling runs: ALIGNN-FF and VASP

B200 runs: NVIDIA B200 GPUs (SLURM partition b200, QOS blackwell_test).

ALIGNN-FF, energy only (no forces), Cu FCC, 1 B200: 58 supercells, $1\times 1\times 1$ to $58\times 58\times 58$ (4 to 780,448 atoms).
Energies correct up to 442,368 atoms ($48\times 48\times 48$); larger sizes are affected by the float32 drift and are shaded red in the table.

VASP, Si diamond: 21 runs over 9 supercell sizes (54 to 8,192 atoms); 19 completed, 2 crashed before the first SCF cycle ($15\times 15\times 15$ and $16\times 16\times 16$).

ALIGNN-FF with energy + forces + stress (added 2026-09-27; the MD workload), Si diamond, 1 GPU, default matpes_r2scan checkpoint (hidden 128, smooth cutoff, 52 neighbours), same script on both cards:
GB10 (121.7 GiB): largest cell 54,872 atoms at 107 GB, OOM at 64,000;
H200 NVL (139.8 GiB): 74,088 atoms at 145 GB, OOM at 85,184.
Memory 1.95 MB/atom on both; the H200 is 4× faster than the GB10 at every size.



ALIGNN-FF results (1/2)

Supercell	Atoms	Neighbor list (s)	Graph (s)	Inference (s)	Total (s)	Model output (eV/atom)
1×1×1	4	0.0844	0.1568	3.2238	3.4650	0.604015
2×2×2	32	0.0015	0.0022	0.0073	0.0111	0.604014
3×3×3	108	0.0018	0.0026	0.0089	0.0132	0.604014
4×4×4	256	0.0087	0.0026	0.0100	0.0212	0.604014
5×5×5	500	0.0035	0.0029	0.0083	0.0147	0.604013
6×6×6	864	0.0047	0.0031	0.0086	0.0165	0.604013
7×7×7	1,372	0.0063	0.0035	0.0096	0.0194	0.604013
8×8×8	2,048	0.0085	0.0040	0.0109	0.0234	0.604013
9×9×9	2,916	0.0115	0.0046	0.0118	0.0279	0.604013
10×10×10	4,000	0.0158	0.0055	0.0136	0.0348	0.604013
11×11×11	5,324	0.0433	0.0181	0.0231	0.0845	0.604013
12×12×12	6,912	0.0680	0.0151	0.0304	0.1135	0.604013
13×13×13	8,788	0.0733	0.0248	0.0476	0.1457	0.604013
14×14×14	10,976	0.0881	0.0216	0.0542	0.1639	0.604013
15×15×15	13,500	0.1020	0.0252	0.0498	0.1771	0.604013
16×16×16	16,384	0.1218	0.0323	0.0682	0.2223	0.604013
17×17×17	19,652	0.1479	0.0399	0.0669	0.2547	0.604013
18×18×18	23,328	0.1598	0.0403	0.0786	0.2787	0.604013
19×19×19	27,436	0.1912	0.0387	0.0928	0.3226	0.604013
20×20×20	32,000	0.2106	0.0414	0.0985	0.3506	0.604013
21×21×21	37,044	0.2482	0.0462	0.1152	0.4096	0.604013
22×22×22	42,592	0.2590	0.0495	0.1283	0.4368	0.604013
23×23×23	48,668	0.2798	0.0548	0.1499	0.4845	0.604013
24×24×24	55,296	0.2096	0.0429	0.1373	0.3897	0.604013
25×25×25	62,500	0.2265	0.0481	0.1548	0.4294	0.604013
26×26×26	70,304	0.2547	0.0537	0.1738	0.4822	0.604015
27×27×27	78,732	0.2841	0.0601	0.1935	0.5377	0.604013
28×28×28	87,808	0.3112	0.0651	0.2156	0.5919	0.604013
29×29×29	97,556	0.3462	0.0750	0.2390	0.6602	0.604013
30×30×30	108,000	0.3773	0.0811	0.2643	0.7227	0.604013

ALIGNN-FF v6 (scale5b_v6.py), one B200, float32. Times are wall time for one energy evaluation.
The first row includes one-time GPU warm-up. Red rows (N > 442,368): the model output is wrong because of the float32 drift; the float64 reference is 0.604015 eV/atom.

ALIGNN-FF results (2/2)

Supercell	Atoms	Neighbor list (s)	Graph (s)	Inference (s)	Total (s)	Model output (eV/atom)
31×31×31	119,164	0.4516	0.0884	0.2913	0.8313	0.604013
32×32×32	131,072	0.4864	0.0953	0.3193	0.9011	0.604013
33×33×33	143,748	0.5267	0.1030	0.3502	0.9799	0.604013
34×34×34	157,216	0.5839	0.1139	0.3820	1.0798	0.604013
35×35×35	171,500	0.6101	0.1259	0.4164	1.1525	0.604013
36×36×36	186,624	0.6554	0.1350	0.4529	1.2434	0.604013
37×37×37	202,612	0.7091	0.1466	0.4912	1.3469	0.604013
38×38×38	219,488	0.7572	0.1677	0.5319	1.4568	0.604013
39×39×39	237,276	0.8124	0.1740	0.5753	1.5617	0.604013
40×40×40	256,000	0.8806	0.1872	0.6202	1.6880	0.604013
41×41×41	275,684	1.0001	0.1986	0.6675	1.8662	0.604013
42×42×42	296,352	1.0695	0.2132	0.7176	2.0003	0.604013
43×43×43	318,028	1.1327	0.2275	0.7691	2.1293	0.604015
44×44×44	340,736	1.1961	0.2483	0.8239	2.2683	0.604013
45×45×45	364,500	1.2776	0.2635	0.8810	2.4221	0.604013
46×46×46	389,344	1.3576	0.2852	0.9407	2.5835	0.604015
47×47×47	415,292	1.4514	0.3117	1.0032	2.7663	0.604015
48×48×48	442,368	1.5614	0.3171	1.0683	2.9467	0.604013
49×49×49	470,596	1.6539	0.3416	1.1366	3.1321	0.606979
50×50×50	500,000	1.7436	0.3695	1.2075	3.3206	0.639071
51×51×51	530,604	1.9523	0.3813	1.2820	3.6156	0.672090
52×52×52	562,432	2.0831	0.4097	1.3590	3.8518	0.702621
53×53×53	595,508	2.1964	0.4336	1.4393	4.0692	0.730894
54×54×54	629,856	2.3269	0.4530	1.5224	4.3024	0.757119
55×55×55	665,500	2.4602	0.4827	1.6091	4.5520	0.781473
56×56×56	702,464	2.5851	0.5138	1.6982	4.7971	0.804124
57×57×57	740,772	2.7414	0.5370	1.7917	5.0702	0.825215
58×58×58	780,448	2.9055	0.5709	1.9083	5.3847	0.844881

ALIGNN-FF v6 (scale5b_v6.py), one B200, float32. Times are wall time for one energy evaluation.
The first row includes one-time GPU warm-up. Red rows (N > 442,368): the model output is wrong because of the float32 drift; the float64 reference is 0.604015 eV/atom.

ALIGNN-FF with forces and stress: GB10 and H200

GPU	Supercell	Atoms	Time (s)	Peak mem (GB)	MB / atom	Model output (eV/atom)
GB10	2×2×2	64	0.043	0.2	3.03	-8.7476
GB10	4×4×4	512	0.114	1.1	2.09	-8.7476
GB10	6×6×6	1,728	0.353	3.4	1.99	-8.7476
GB10	8×8×8	4,096	0.635	8.1	1.97	-8.7475
GB10	10×10×10	8,000	1.174	15.7	1.96	-8.7479
GB10	12×12×12	13,824	1.959	27.0	1.96	-8.7479
GB10	14×14×14	21,952	3.116	42.9	1.95	-8.7476
GB10	15×15×15	27,000	3.742	52.7	1.95	-8.7470
GB10	16×16×16	32,768	4.576	64.0	1.95	-8.7464
GB10	17×17×17	39,304	5.504	76.8	1.95	-8.7467
GB10	18×18×18	46,656	6.454	91.1	1.95	-8.7470
GB10	19×19×19	54,872	7.592	107.1	1.95	-8.7475
GB10	20×20×20	64,000	OOM	-	-	-
H200	2×2×2	64	0.020	0.2	3.05	-8.7476
H200	4×4×4	512	0.032	1.1	2.09	-8.7476
H200	6×6×6	1,728	0.075	3.4	1.99	-8.7476
H200	8×8×8	4,096	0.159	8.1	1.97	-8.7475
H200	10×10×10	8,000	0.278	15.7	1.96	-8.7479
H200	12×12×12	13,824	0.470	27.0	1.96	-8.7479
H200	14×14×14	21,952	0.724	42.9	1.95	-8.7476
H200	15×15×15	27,000	0.905	52.7	1.95	-8.7470
H200	16×16×16	32,768	1.175	64.0	1.95	-8.7464
H200	17×17×17	39,304	1.339	76.8	1.95	-8.7467
H200	18×18×18	46,656	1.578	91.1	1.95	-8.7470
H200	19×19×19	54,872	1.887	107.1	1.95	-8.7475
H200	20×20×20	64,000	2.184	124.9	1.95	-8.7482
H200	21×21×21	74,088	2.534	144.6	1.95	-8.7482
H200	22×22×22	85,184	OOM	-	-	-

ALIGNN-FF energy + forces + stress single points (bench_ff_scaling.py, default matpes_r2scan checkpoint, md5 92cfe295), Si diamond 8-atom conventional cell repeated $i \times i \times i$, float32. Time = median of three calls including the graph build; memory = peak allocated on the GPU. Red row = the first size that did not fit. GB10 = atomgptlab job 13474; H200 NVL = Skipjack job 913733 (node gh204). Both 2026-09-27.

VASP results

Supercell	Atoms	GPUs	k-points	Bands	SCF cycles	Elapsed (s)	s / SCF	E (eV/atom)	Status
3×3×3	54	1	3	136	20/20	15.3	0.76	-5.4105	converged
3×3×3	54	2	3	136	20/20	17.8	0.89	-5.4105	converged
3×3×3	54	4	3	136	20/20	18.3	0.91	-5.4105	converged
3×3×3	54	8	3	136	20/20	27.3	1.36	-5.4105	converged
4×4×4	128	1	3	321	20/20	39.6	1.98	-5.4211	converged
4×4×4	128	2	3	321	20/20	38.3	1.92	-5.4211	converged
4×4×4	128	4	3	321	20/20	36.5	1.82	-5.4211	converged
4×4×4	128	8	3	321	20/20	42.5	2.13	-5.4211	converged
5×5×5	250	1	1	626	20/20	57.6	2.88	-5.3903	converged
5×5×5	250	2	1	626	20/20	62.4	3.12	-5.3903	converged
5×5×5	250	4	1	628	20/20	75.3	3.76	-5.3903	converged
5×5×5	250	8	1	632	20/20	108.1	5.41	-5.3903	converged
6×6×6	432	1	1	1,081	20/20	113.7	5.68	-5.4111	converged
6×6×6	432	2	1	1,082	20/20	96.2	4.81	-5.4111	converged
6×6×6	432	4	1	1,084	20/20	89.3	4.46	-5.4111	converged
6×6×6	432	8	1	1,088	20/20	123.6	6.18	-5.4111	converged
10×10×10	2,000	4	1	5,004	5/5	518.4	103.69	-5.5124	not converged
12×12×12	3,456	8	1	8,648	2/2	678.6	339.29	-3.1397	not converged
14×14×14	5,488	8	1	13,728	2/2	1561.4	780.69	-3.1433	not converged
15×15×15	6,750	8	1	16,880	0/2	-	-	-	crashed
16×16×16	8,192	8	1	20,488	0/2	-	-	-	crashed

VASP GPU build, single-point SCF. 3×3×3 to 6×6×6: NELM=20, EDIFF=1E-6. 10×10×10: NELM=5, ALGO=Fast. 12×12×12 and larger: NELM=2, ALGO=Fast. SCF cycles = completed / maximum allowed (NELM). E = energy of the last SCF cycle / atom count. Converged = last energy change below the run's EDIFF. Amber rows were stopped early for timing; their E is an unconverged intermediate value, not a result. Red rows crashed before the first SCF cycle (likely out of memory).